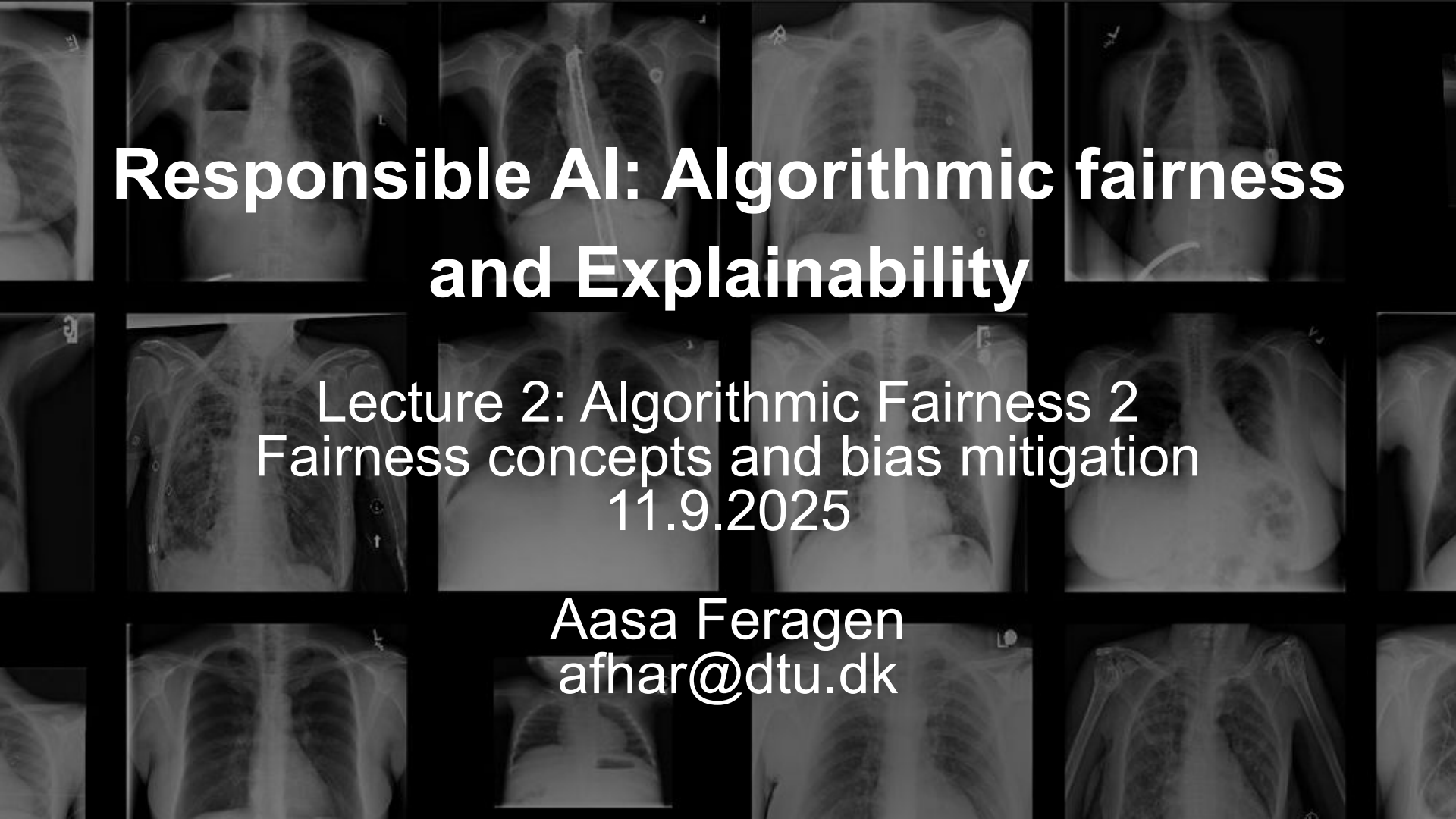


Responsible AI: Algorithmic fairness and Explainability

Lecture 2: Algorithmic Fairness 2
Fairness concepts and bias mitigation
11.9.2025

Aasa Feragen
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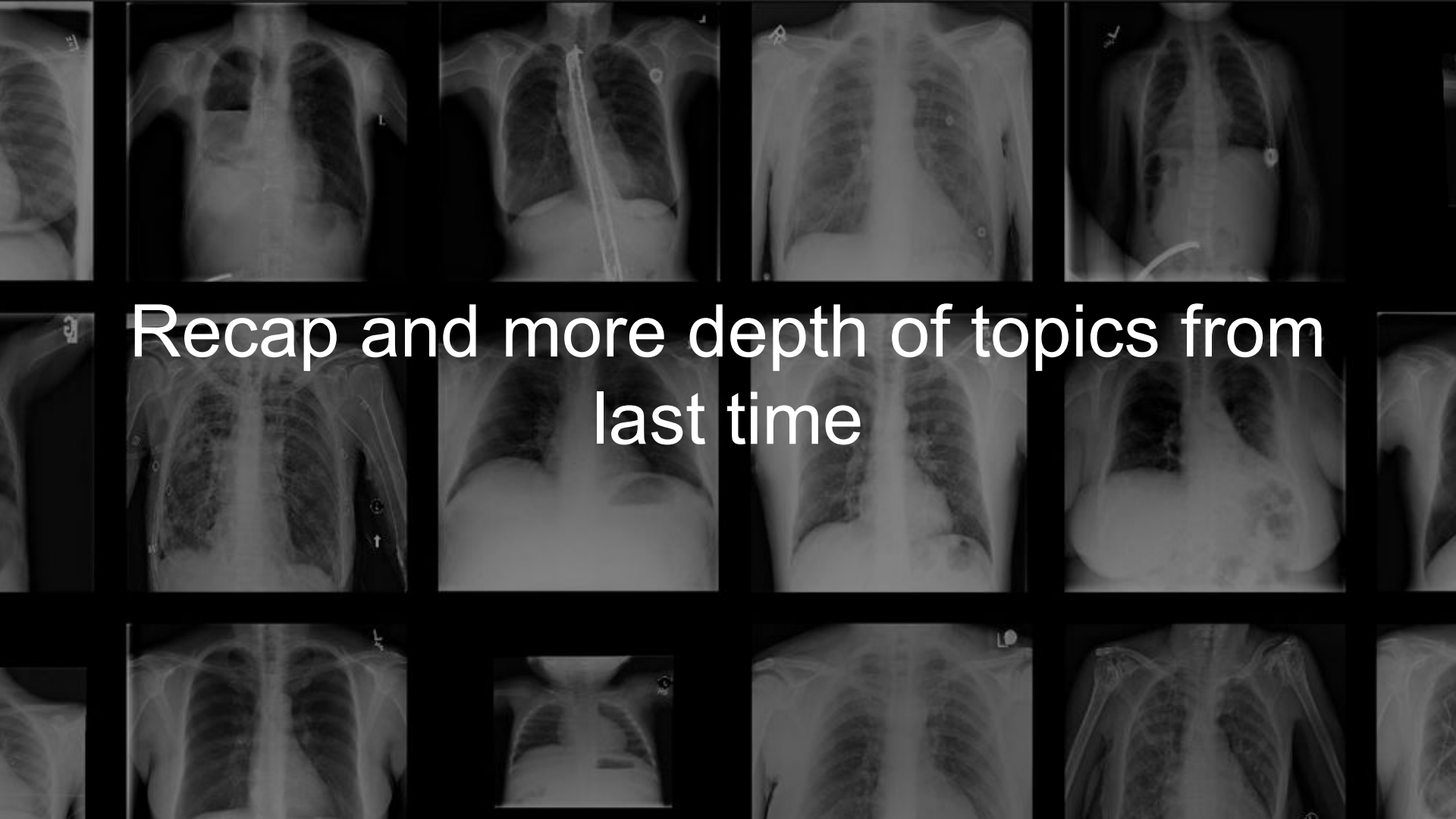
By the end of this lecture, you should:

- Be more closely familiar the three main algorithmic fairness concepts from last time: Independence, Sufficiency, and Separation

- Know the three different paradigms of fairness mitigation:
  - Preprocessing
  - In-training
  - Postprocessing

  and some examples from each type

- In particular, you should be closely familiar with *randomization* which is the method used by the Threshold Optimizer in FairLearn



Recap and more depth of topics from last time

Three main fairness concepts

Three different categories of "fairness"

- Independence
- Separation (Equalized odds)
- Sufficiency

These correspond to equalizing across groups

- Acceptance rate $\mathbb{P}\{\hat{Y} = 1\}$ of the classifier
- The error rates $\mathbb{P}\{\hat{Y} = 0 | Y = 1\}$ and $\mathbb{P}\{\hat{Y} = 1 | Y = 0\}$ of the classifier
- Outcome frequency $\mathbb{P}\{Y = 1 | R = r\}$ of a score R

Independence

- **Definition:**
  The model R satisfies *independence* if $R \perp A$.
- Applied to binary predictions $\hat{Y}$, this is equivalent to

$$\mathbb{P}\{\hat{Y} = 1 | A = a\} = \mathbb{P}\{\hat{Y} = 1 | A = b\}$$

  for all groups a, b
- That is, equalizing the acceptance rate $\mathbb{P}\{\hat{Y} = 1\}$ of the classifier across groups

Question

In which ways is this a *good* criterion? In which ways is it *bad*? E.g. consider scenarios:

- Hiring female or male software developers
- Diagnosing cancer



Separation (Equalized Odds)

- **Definition:**
  The model R satisfies *separation* if $R \perp A | Y$.
- **Intuition:** The target Y quantifies underlying *merit*
- Applied to binary predictions $\hat{Y}$, this is becomes:

$$\mathbb{P}\{\hat{Y} = 1 | Y = 1, A = a\} = \mathbb{P}\{\hat{Y} = 1 | Y = 1, A = b\}$$

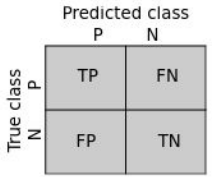
$$\mathbb{P}\{\hat{Y} = 1 | Y = 0, A = a\} = \mathbb{P}\{\hat{Y} = 1 | Y = 0, A = b\}$$

- Equalizing *true positive rate* and *false positive rate*
- Equivalent: Equalizing *false negative rate* and *true negative rate*

Question

In which ways is this a *good* or *bad* criterion? E.g.:

- Hiring female and male software developers
- Diagnosing cancer



Sufficiency

- The model R satisfies *sufficiency* if $Y \perp A | R$
- That is, for all groups a, b and all r, we have

$$\mathbb{P}\{Y = 1 | R = r, A = a\} = \mathbb{P}\{Y = 1 | R = r, A = b\}$$

- For binary classifiers $f(X) = \hat{Y} \in \{0, 1\}$, this indicates equal positive and negative predictive values

$$\mathbb{P}\{Y = 1 | \hat{Y} = 1, A = a\} = \mathbb{P}\{Y = 1 | \hat{Y} = 1, A = b\}$$

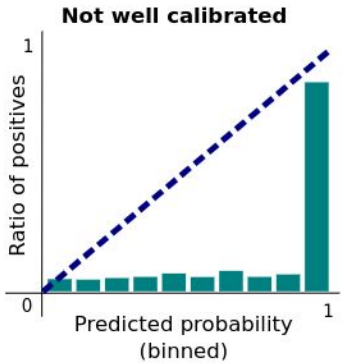
$$\mathbb{P}\{Y = 0 | \hat{Y} = 0, A = a\} = \mathbb{P}\{Y = 0 | \hat{Y} = 0, A = b\}$$

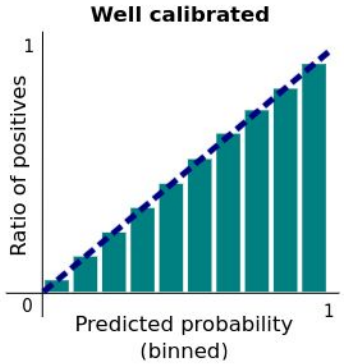
Question

In which ways is this a *good* criterion? In which ways is it *bad*?

How sufficiency relates to calibration

- Sufficiency – for probabilistic classifiers – corresponds to equalizing the outcome frequency $\mathbb{P}\{Y = 1|R = r\}$ of the score R across groups A for all scores $r \in R$.
- This is closely related to *calibration*.



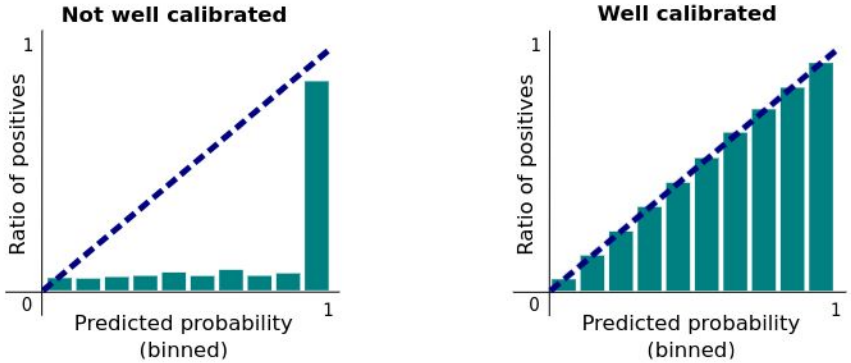


How sufficiency relates to calibration

- Sufficiency – for probabilistic classifiers – corresponds to equalizing the outcome frequency $\mathbb{P}\{Y = 1|R = r\}$ of the score R across groups A for all scores $r \in R$.
- This is closely related to *calibration*.
- A model R is *calibrated* with respect to the target Y if for all scores $r \in [0, 1]$, we have

$$\mathbb{P}\{Y = 1|R = r\} = r.$$

- Equivalently: on a population level, the predicted foreground probability equals the empirical one.



How sufficiency relates to calibration

Why is this interesting?

- Sufficiency:
  $\mathbb{P}\{Y = 1 | R = r, A = a\} = \mathbb{P}\{Y = 1 | R = r, A = b\}$
- Calibration: $\mathbb{P}\{Y = 1 | R = r\} = r$

Theorem

- *Calibration by group implies sufficiency*
- *If a model R satisfies sufficiency, then there exists a calibrating function l: $[0, 1] \to [0, 1]$ so that $l(R)$ satisfies calibration by group.*

Calibration is well studied and we have off-the-shelf tools to calibrate models. This means we can obtain sufficiency through calibration.

Diagnostics based on algorithmic fairness criteria

Recall: The three categories of "fairness" correspond to equalizing across groups

- **Independence:** Acceptance rate $\mathbb{P}\{\hat{Y} = 1\}$ of the classifier
- **Separation:** The error rates $\mathbb{P}\{\hat{Y} = 0 | Y = 1\}$ and $\mathbb{P}\{\hat{Y} = 1 | Y = 0\}$ of the classifier
- **Sufficiency:** Outcome frequency $\mathbb{P}\{Y = 1 | R = r\}$ of a score R

Diagnosing bias is easy! A simple diagnostic of "unfairness" can be derived by looking at either difference or fractions of the associated rates or frequencies between groups.

Tools

A number of software tools are freely available, of different quality and usability. Let's see an example from the Fairlearn package, which offers easy bias diagnostics:

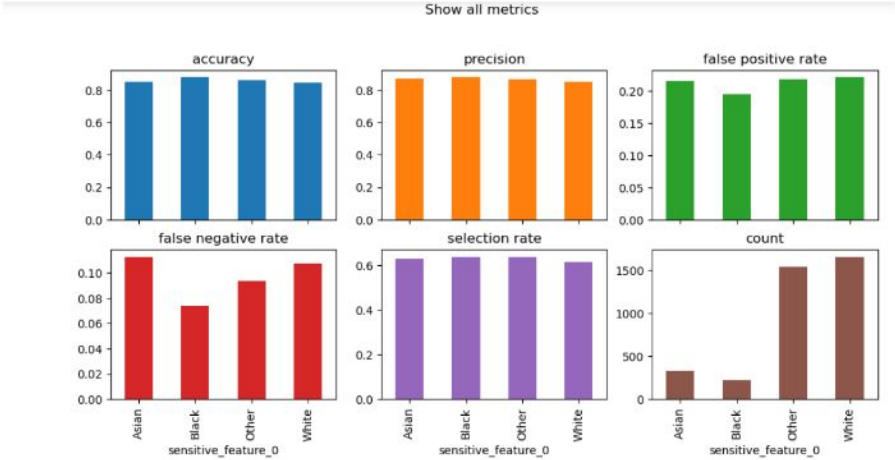


Figure: Fairness diagnostics with Fairlearn

Irvin, Jeremy, et al. AAAI'19

Bird Sarah, et al. Microsoft, Tech. Rep. MSR-TR-2020-32 (2020).


How do we make models more fair?
“Bias mitigation”

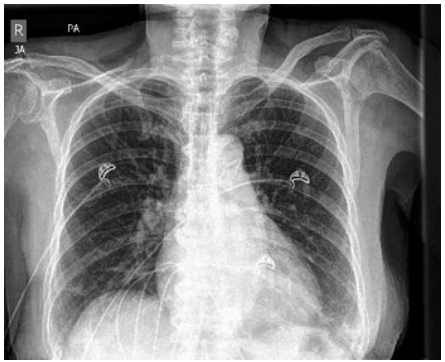
Bias mitigation – How to obtain a fair classifier?

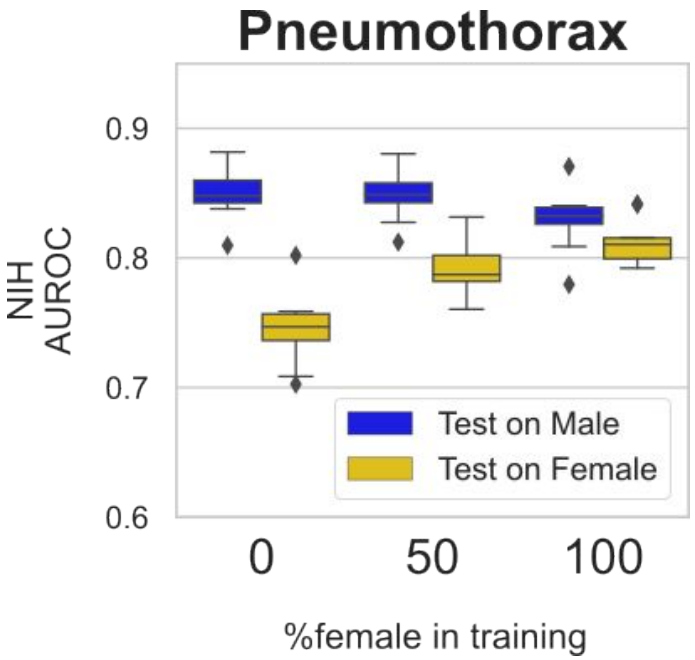
Classically, we group “bias mitigation” approaches into three different types

- **Preprocessing:**
  Align or process the data beforehand to remove potential biases
- **In-training:**
  Constrain the training of the algorithm so as to only obtain fair classifiers (given a definition of “fair”)
- **Postprocessing:**
  Adapt the predictions made by the model so as to make them fair

Bias mitigation through preprocessing

- Mitigating group imbalance via more data, upsampling or augmentation
- Within-group standardization or harmonization
- **If possible:** Standardizing data quality across groups





Weng et al, FAIMI'23; reproduced from Larrazabal et al, PNAS'20

Do you see any risks in bias mitigation via data preprocessing?

Bias mitigation through preprocessing - nothing is simple

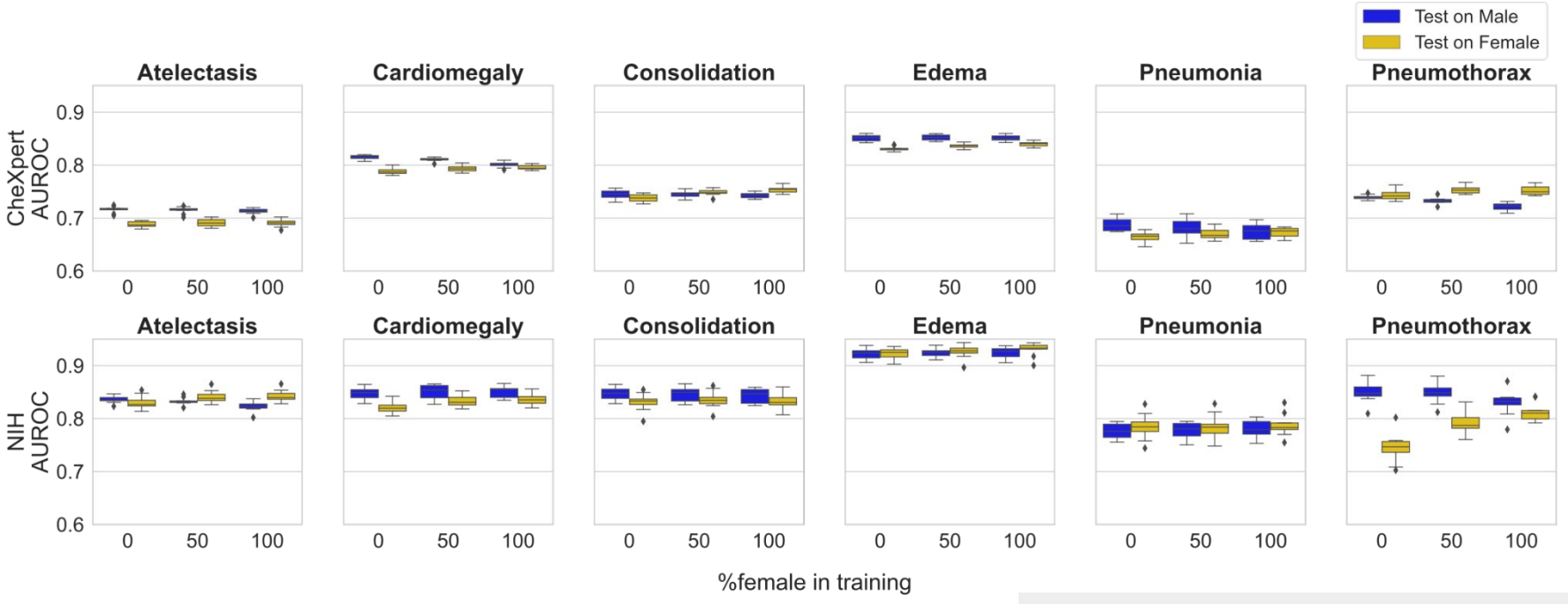


Figure by Weng et al, FAIMI'23; reproduced from Larrazabal et al, PNAS'20.

In-training mitigation

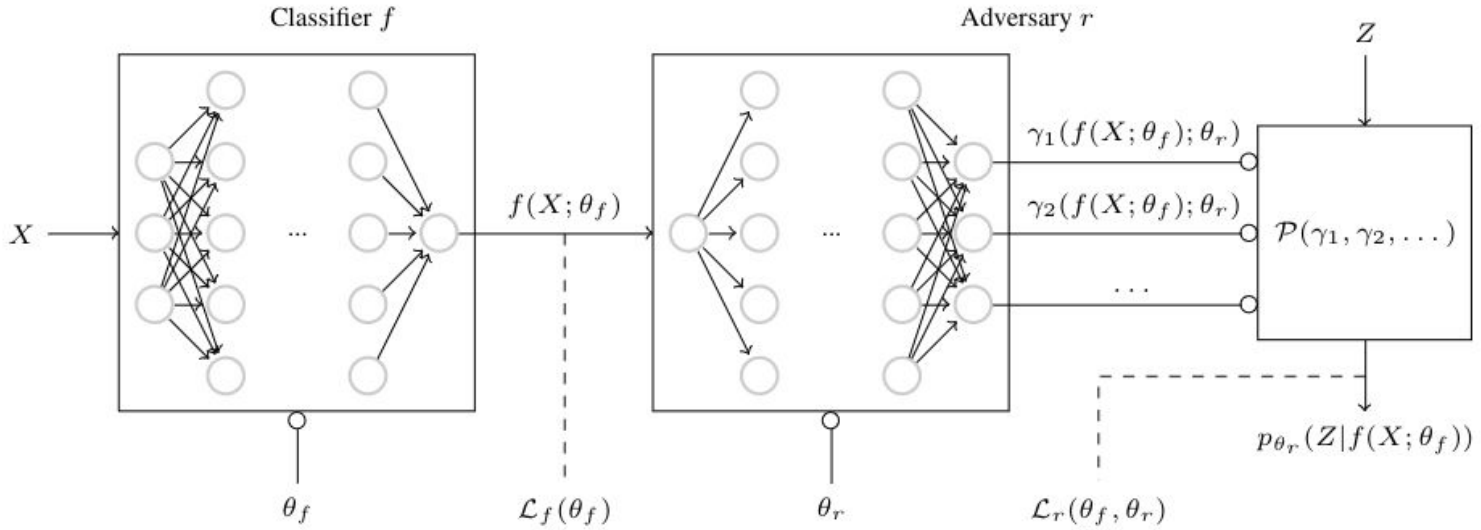
Example approaches include:

- Adversarial training to avoid encoding of sensitive attributes
- Adding terms to the loss function that encourage a particular fairness criterion
- Using causal modelling to eliminate unwanted causal dependencies in model

In-training mitigation

Example approaches include:

- Adversarial training to avoid encoding of sensitive attributes
- erion
- odel



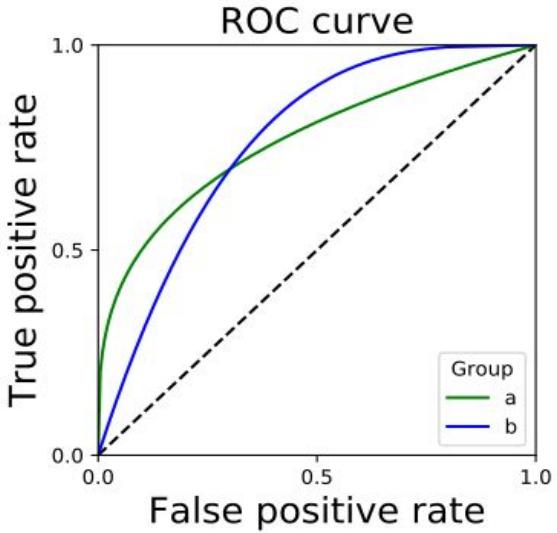
Louppe et al, Learning to pivot with Adversarial Networks, NeurIPS 2017

Do you see any risks associated with in-training bias mitigation?

Post-hoc mitigation – massage a biased model to make it less biased

- **ROC curves:** Often used to understand classifier performance
- **AUC:** Area under the ROC curve – closer to 1 for a better and more robust classifier

- Is the green or blue ROC curve better?
- Is the corresponding classifier fair?



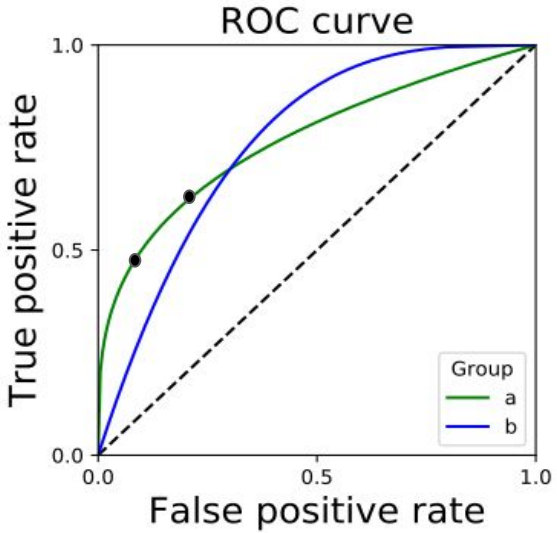
Post-hoc mitigation – massage a biased model to make it less biased

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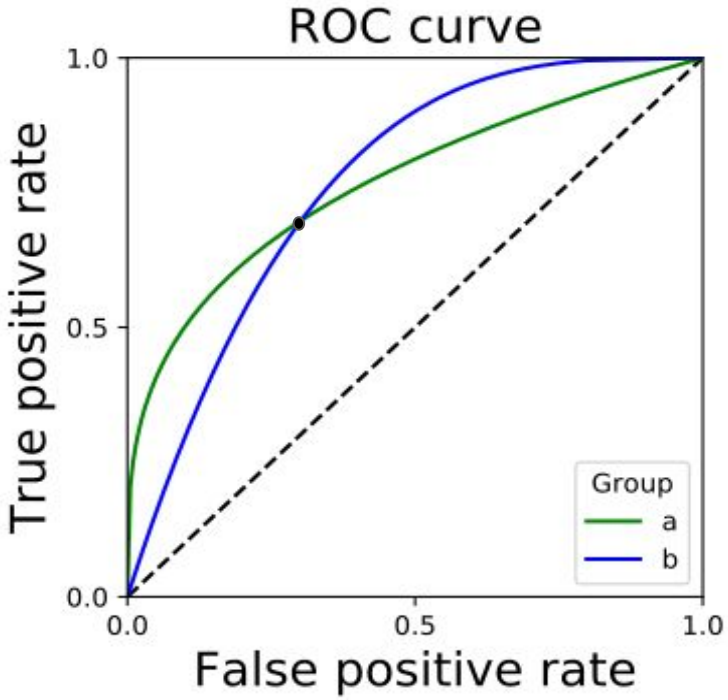
- If the dots are at threshold 0.5, is the thresholded classifier fair?

- Can it become fair?

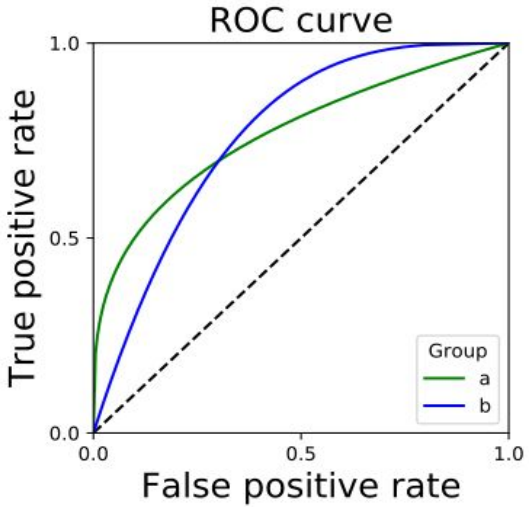


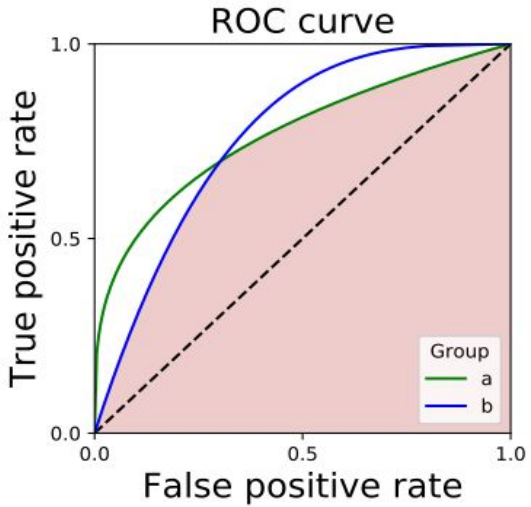
Post-hoc mitigation – massage a biased model to make it less biased

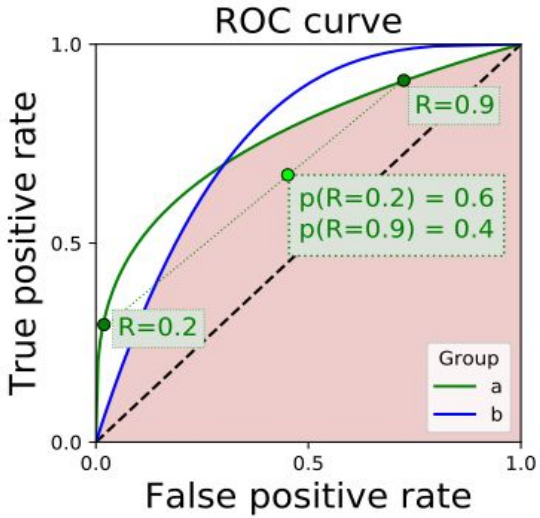
- **Simple solution!** Pick the intersection!
- Is this a good idea?



Better idea? Equalized odds by randomization (balancing TPR + FPR)



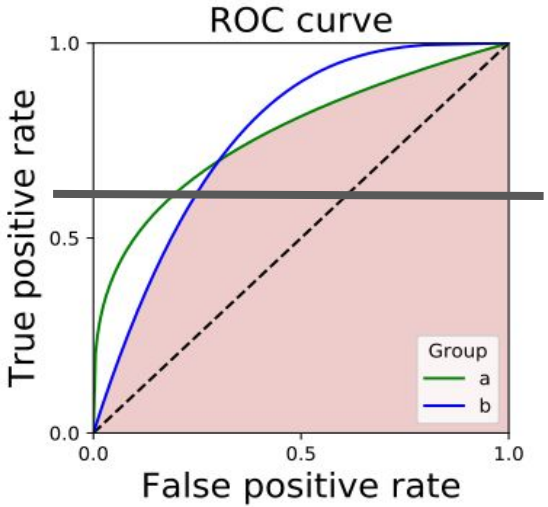


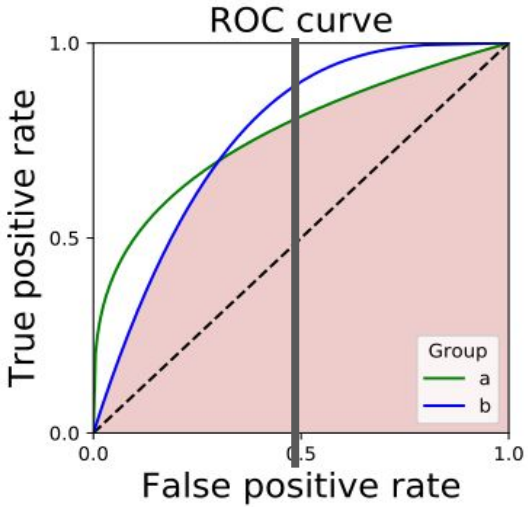


TPR = TP/Actual positive; FPR = FP/True negative

Randomization (Hardt et al, NeurIPS 2016) interpolates TPR/FPR by sampling pairs of thresholds according to an interpolant probability p. This is the method of “Threshold optimizer” technique in FairLearn.

Relaxation? Equal opportunity considering either TPR or FPR, not both.





TPR = TP/Actual positive; FPR = FP/True negative

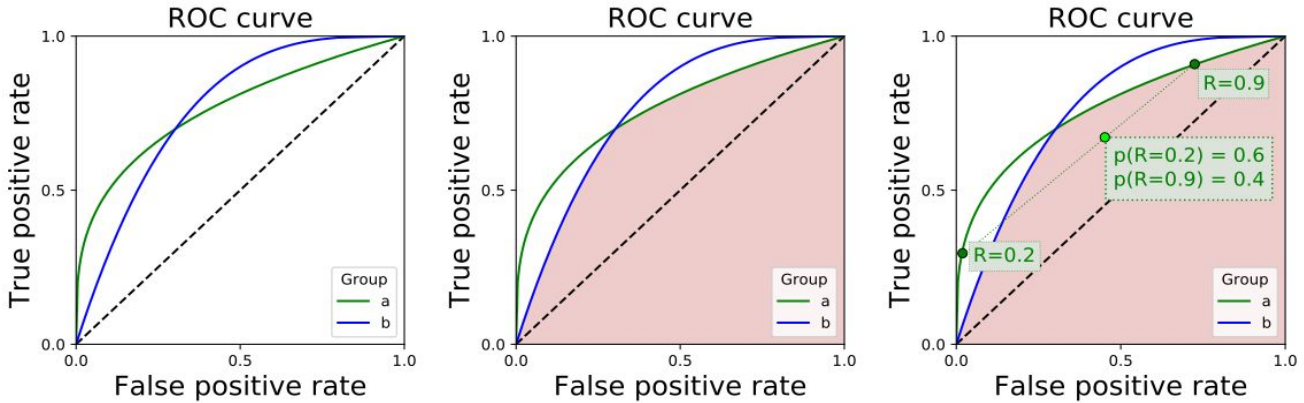
Randomization (Hardt et al, NeurIPS 2016) interpolates TPR/FPR by sampling pairs of thresholds according to an interpolant probability p. This is the method of “Threshold optimizer” technique in FairLearn.

Post-hoc mitigation – summary

We have discussed the following postprocessing mitigation methods:

- Equalized odds by randomization (balancing TPR + FPR)
- Relaxation of equalized odds: Equal opportunity (balancing TPR only)
- Group-wise threshold selection for other criteria

Randomization (Hardt et al, NeurIPS 2016) interpolates TPR/FPR by sampling pairs of thresholds according to an interpolant probability p. This underlies the “Threshold optimizer” technique from the Fairlearn package.

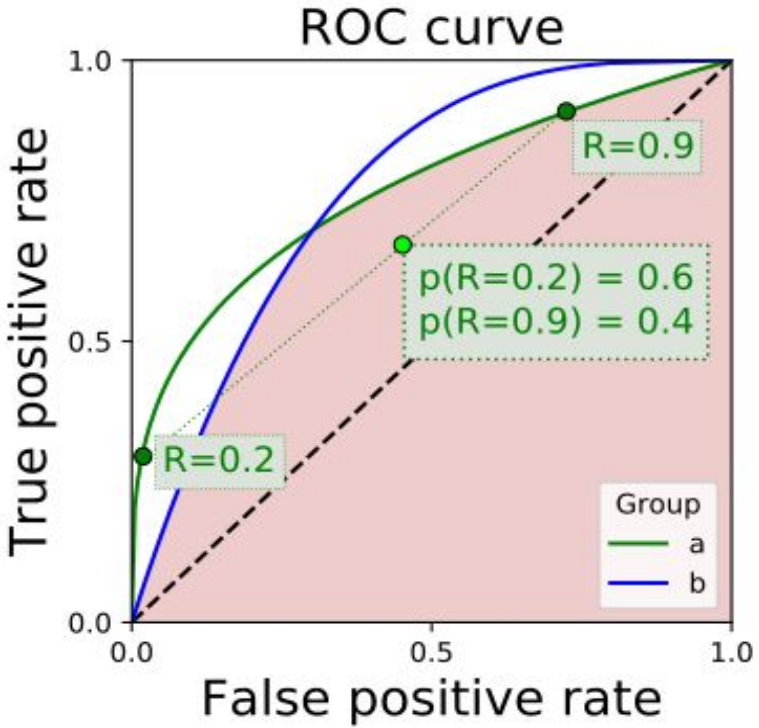


Do you see any risks associated with post-processing bias mitigation?

The fairness-performance tradeoff

Enforcing fairness can come with a performance drop

- Randomization and in-training techniques are sensitive
- **Alternative:** Know your bias (diagnostics), use suitable generalization tools to minimize it, and report what is left



By now, you should:

- Be closely familiar with the three main notions of algorithmic fairness: Independence, Separation and Sufficiency
- Be closely familiar with different ways to mitigate bias, along with their inherent risks

Next time: A paper discussion exercise

- You will get a short videolecture to watch before class
- Additionally, you will be assigned three papers to read
- In class, you will discuss the papers first in groups, and then with the TAs
- Finally, you will get a reasonably short followup videolecture to watch after the class, which will support your project